

A partial answer to the non-reflexive operator-algebra question in
arXiv:1707.08399

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Source question

Beanland, Kania and Laustsen ask in Section 3 of *The algebras of bounded operators on the Tsirelson and Baernstein spaces are not Grothendieck spaces*, arXiv:1707.08399:

Is there a non-reflexive Banach space X for which $\mathcal{B}(X)$ is a Grothendieck space?
If such a space X exists, both X and X^* would necessarily be Grothendieck spaces,
because $\mathcal{B}(X)$ contains complemented copies of them.

Pfützner [14] showed that C^* -algebras have Pełczyński's property (V). This implies that von Neumann algebras are Grothendieck spaces, so in particular $\mathcal{B}(H)$ is a Grothendieck space for each Hilbert space H . No other examples of infinite-dimensional Banach spaces X for which $\mathcal{B}(X)$ is a Grothendieck space are known. In particular, the following questions remain open:

- Is $\mathcal{B}(\ell_p)$ a Grothendieck space for $p \in (1, \infty) \setminus \{2\}$? In this case, the problem is equivalent to deciding whether $(\bigoplus_{n \in \mathbb{N}} \mathcal{B}(\ell_p^n))_{\ell_\infty}$ is a Grothendieck space because $\mathcal{B}(\ell_p)$ and $(\bigoplus_{n \in \mathbb{N}} \mathcal{B}(\ell_p^n))_{\ell_\infty}$ are isomorphic as Banach spaces (see [1, Theorem 2.1]; for $p = 2$, see also [7, p. 317], where the result is credited to Haagerup and Lindenstrauss).
- Is $\mathcal{B}(X)$ a Grothendieck space for every weak Hilbert space X ?
- More generally, is $\mathcal{B}(X)$ a Grothendieck space for every super-reflexive Banach space X ?
- Let X be a reflexive Banach space with an unconditional finite-dimensional Schauder decomposition $(F_n)_{n=1}^\infty$ whose blocks F_n are uniformly isomorphic to $\ell_1^{m_n}$, where $m_n = \dim F_n \rightarrow \infty$ as $n \rightarrow \infty$. Is it true that $\mathcal{B}(X)$ is not a Grothendieck space? If it is true, it would provide a unified proof of the two cases considered in Theorem 1.1.
- Is there a non-reflexive Banach space X for which $\mathcal{B}(X)$ is a Grothendieck space? If such a space X exists, both X and X^* would necessarily be Grothendieck spaces (because $\mathcal{B}(X)$ contains complemented copies of them), and therefore X would be non-separable. No non-reflexive Grothendieck space X for which X^* is also a Grothendieck space is known.

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The result below rules out all infinite-dimensional spaces of the form $C(K)$.

Result

Theorem 1. *Let K be an infinite compact Hausdorff space. Then the Banach space $\mathcal{B}(C(K))$ is not Grothendieck.*

Corollary 1. *No infinite-dimensional $C(K)$ space can be a positive example for the non-reflexive question in arXiv:1707.08399.*

Proof

We use two standard permanence facts, included for completeness.

Lemma 1. *If a Banach space Z is Grothendieck, then every complemented subspace of Z is Grothendieck.*

Proof. Let Y be complemented in Z , and let $P : Z \rightarrow Y$ be a bounded projection. Suppose (y_n^*) is a weak-star null sequence in Y^* . Then $z_n^* = y_n^* \circ P$ is weak-star null in Z^* . If Z is Grothendieck, then (z_n^*) is weakly null in Z^* . For $y^{**} \in Y^{**}$ define $z^{**} \in Z^{**}$ by $z^{**}(z^*) = y^{**}(z^*|_Y)$. Since $z^{**}(z_n^*) = y^{**}(y_n^*)$, the sequence (y_n^*) is weakly null. $\square$

Lemma 2. *The space ℓ_1 is not Grothendieck.*

Proof. A separable Grothendieck space is reflexive: if X is separable, the dual unit ball B_{X^*} is weak-star compact metrizable, so every sequence in it has a weak-star convergent subsequence. If X is Grothendieck, that subsequence is weakly convergent. Hence B_{X^*} is weakly sequentially compact, and the Eberlein–Smulian theorem makes it weakly compact. Thus X^* , and therefore X , is reflexive. Since ℓ_1 is separable and not reflexive, it is not Grothendieck. $\square$

Now let K be infinite. Choose a countably infinite subset $D = \{t_1, t_2, \dots\}$ of K . Under the Riesz representation identification $C(K)^* = M(K)$, the closed subspace

$$A_D = \left\{ \sum_{j=1}^{\infty} a_j \delta_{t_j} : (a_j) \in \ell_1 \right\}$$

is isometric to $\ell_1(D)$. It is also 1-complemented in $M(K)$. Indeed, for $\mu \in M(K)$ put

$$P_D \mu = \sum_{j=1}^{\infty} \mu(\{t_j\}) \delta_{t_j}.$$

The total variation of this atomic part is bounded by $\|\mu\|$, so P_D is a contractive projection from $M(K)$ onto A_D .

Since A_D is isometric to ℓ_1 and ℓ_1 is not Grothendieck, the first permanence lemma shows that $M(K) = C(K)^*$ is not Grothendieck.

Finally, $C(K)^*$ is complemented in $\mathcal{B}(C(K))$ by rank-one operators. Fix $t_0 \in K$, let $\mathbf{1}$ denote the constant-one function, and define

$$J : M(K) \rightarrow \mathcal{B}(C(K)), \quad J(\mu)(f) = \mu(f)\mathbf{1}.$$

Define also

$$Q : \mathcal{B}(C(K)) \rightarrow M(K), \quad Q(T)(f) = \delta_{t_0}(Tf).$$

Then $QJ = \text{id}_{M(K)}$, so $J(M(K))$ is complemented in $\mathcal{B}(C(K))$. If $\mathcal{B}(C(K))$ were Grothendieck, then $M(K)$ would be Grothendieck by the first lemma, contradicting the preceding paragraph. Therefore $\mathcal{B}(C(K))$ is not Grothendieck.

Verification and limitations

The proof uses only complemented-subspace permanence of the Grothendieck property, the elementary fact that separable Grothendieck spaces are reflexive, the Riesz representation theorem for $C(K)^*$, and explicit rank-one embeddings. No computational verification is required.

This is a partial result. It excludes the classical non-reflexive family $C(K)$, K infinite, but does not solve the source’s general non-reflexive problem. In particular, it does not rule out a nonseparable Banach space X for which both X and X^* are Grothendieck, and it does not address the source’s questions about $\mathcal{B}(\ell_p)$, weak Hilbert spaces, super-reflexive spaces, or reflexive spaces with unconditional FDDs.

Novelty check

The local run indexes and existing attempts for arXiv:1707.08399 were checked on 2026-06-29; no existing packet recorded this $C(K)$ exclusion. Bounded web searches for the exact operator-algebra questions around $\mathcal{B}(\ell_p)$ and Grothendieck operator algebras did not locate a later exact resolution of the source's open-problem list. Because the present argument is short and uses standard facts, it should be reviewed as a likely-valid partial exclusion with possible folklore status rather than as a claimed full solution.

References

- [1] K. Beanland, T. Kania and N. J. Laustsen, *The algebras of bounded operators on the Tsirelson and Baernstein spaces are not Grothendieck spaces*, arXiv:1707.08399.